

Direct two-root endpoint Hessian and uniform local source gap

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-08-03 · this version issued 2026-08-03 · v1.0

1. Purpose and scope: result and boundary

R-150 installed the actual A1 antipodal field and current synthesis but left the relative final secant and earlier-root feedback open. The subsequent matrix route tried to manufacture separate forest, balanced, and low blocks. That is unnecessary and unsafe: after the terminal scalar owner is fixed, those objects are alternate expansions or estimates of the same owner, not new orthogonal reserves. The correct finite-cylinder matrix is obtained by differentiating the complete A7 endpoint once.

This note carries out that direct differentiation for the full six-real, both-sign, linear adapted chart

$$h_2 = tH\xi_1, \quad H \in \mathbb{R}^{12 \times 12}, \quad (1.1)$$

where ξ_1 is the simultaneous $\{\pm p\}$ root and h_2 shifts the simultaneous $\{\pm 2p\}$ root. Every sign-to-sign and internal-component coupling is retained, so the Hessian has 144 real coordinates. Here p is an admissible nonzero dual-lattice momentum and the finite cutoff retains all four modes $\pm p, \pm 2p$.

The exact endpoint Hessian is reduced to one covariance response tensor. The global R-130 pointwise coefficient-Hessian envelope then proves, for every such retained pair and every stationary common-even Gaussian background,

$$\mathcal{R}''(0)[H, H] > -\frac{4}{5}\|H\|_{\text{HS}}^2, \quad H \neq 0. \quad (1.2)$$

The Cameron–Martin term $(9/20)\mathbb{E}\|h_2\|^2$ has Hessian $9\|H\|_{\text{HS}}^2/10$, while the terminal sixth-power Hessian is nonnegative. Consequently the augmented local Hessian has the strict certified gap

$$\boxed{\mathcal{A}''(0)[H, H] > \frac{1}{10}\|H\|_{\text{HS}}^2, \quad H \neq 0.} \quad (1.3)$$

The radial estimate is uniform for every $x = |p|^2 \geq 0$; the torus theorem applies only to admissible nonzero dual-lattice p for which both roots are retained. It is not the nonlinear, nonzero-past, multi-root aggregation required by T-050. It neither identifies the historical complete-low object nor selects a spatial phase or a PDE.

2. Production chart and covariance paths

The A1 complex symbol is

$$A(p) = \{r + Z|p|^2 + Y|p|^4\}\mathbf{I}_3 + M, \quad (2.1)$$

with

$$M = \begin{pmatrix} 1/10 & -1/20 & -1/20 \\ -1/20 & 13/100 & -1/20 \\ -1/20 & -1/20 & 17/100 \end{pmatrix}. \quad (2.2)$$

The A6/A7 convention gives one directed six-real covariance $\Gamma(p) = \text{diag}(A(p)^{-1}, A(p)^{-1})$. Let $G_p G_p^* = \Gamma(p)$, $J = \begin{pmatrix} 0 & -I_3 \\ I_3 & 0 \end{pmatrix}$, and $U(\theta) = \exp(\theta J)$. On the torus of volume V define

$$S_1(x) = \frac{1}{\sqrt{V}} \begin{bmatrix} U(p \cdot x) G_p & U(-p \cdot x) G_p \end{bmatrix}, \quad (2.3)$$

$$S_2(x) = \frac{1}{\sqrt{V}} \begin{bmatrix} U(2p \cdot x) G_{2p} & U(-2p \cdot x) G_{2p} \end{bmatrix}. \quad (2.4)$$

For $i = 1, 2, 3$, let $D_{r,i} = \partial_i S_r$ and write D_r for their Hilbert direct sum. Their norms obey

$$\|D_1\|_{\text{op}} = |p| \|S_1\|_{\text{op}}, \quad \|D_2\|_{\text{op}} = 2|p| \|S_2\|_{\text{op}}. \quad (2.5)$$

Bounded common-even scalar regulator multipliers of modulus at most one can only decrease these norms.

Let all uncontrolled modes be included in a stationary common-even Gaussian background. At one spatial point write the controlled endpoint as

$$W_t = (S_1 + t S_2 H) \xi_1 + S_2 \xi_2 + W_{\text{bg}}, \quad V_{t,i} = (D_{1,i} + t D_{2,i} H) \xi_1 + D_{2,i} \xi_2 + V_{\text{bg},i}. \quad (2.6)$$

The background may contain the full sharp cube. Simultaneous antipodal revelation and common evenness give

$$K_0(x) := \text{Cov}(W_0(x), V_0(x)) = 0. \quad (2.7)$$

Put $C_t = \text{Cov}(W_t)$, $Q_t = \sum_i \text{Cov}(V_{t,i})$, and $K_{t,i} = \text{Cov}(W_t, V_{t,i})$. Their relevant coefficients are

$$\begin{aligned} C_1 &= S_2 H S_1^* + S_1 H^* S_2^*, & C_2 &= S_2 H H^* S_2^*, \\ Q_1 &= \sum_i (D_{2,i} H D_{1,i}^* + D_{1,i} H^* D_{2,i}^*), & Q_2 &= \sum_i D_{2,i} H H^* D_{2,i}^*, \\ K_{1,i} &= S_2 H D_{1,i}^* + S_1 H^* D_{2,i}^*. \end{aligned} \quad (2.8)$$

3. One scalar endpoint and its exact Hessian

Let $B(w) = C_6(w)^* C_6(w) \succeq 0$ be the exact six-row A7 coefficient matrix and let Q_0 be the primitive derivative covariance used in the normal ordering. The baseline-subtracted endpoint is

$$\mathcal{R}(t) = \frac{1}{2} \int \mathbb{E} \left\{ \sum_i V_{t,i}^* B(W_t) V_{t,i} - \text{Tr}[B(W_t) Q_0] \right\} dx. \quad (3.1)$$

For a centered joint Gaussian pair, two Gaussian integrations by parts give the exact identity

$$\begin{aligned} 2\mathcal{R}(t) &= \int \{ \text{Tr}[(Q_t - Q_0) \bar{B}(C_t)] \\ &\quad + \sum_i (K_{t,i})_{am} (K_{t,i})_{bn} \bar{T}_{abmn}(C_t) \} dx, \end{aligned} \quad (3.2)$$

where

$$\bar{B}(C) = \mathbb{E}_C B(W), \quad \bar{T}_{abmn}(C) = \mathbb{E}_C \partial_{ab} B_{mn}(W). \quad (3.3)$$

Equivalently, when C is invertible,

$$\bar{T}_{abmn} = \mathbb{E} \left[\{ (C^{-1} W)_a (C^{-1} W)_b - (C^{-1})_{ab} \} B_{mn}(W) \right]. \quad (3.4)$$

This derivative-free score formula is useful for numerical audit but is not needed for the rigorous bound below.

Since $K_0 = 0$, differentiating (3.2) gives

$$\boxed{\mathcal{R}''(0)[H, H] = \int \left\{ \text{Tr}(Q_2 \bar{B}) + \frac{1}{2}(Q_1)_{mn}(C_1)_{ab} \bar{T}_{abmn} + \sum_i (K_{1,i})_{am} (K_{1,i})_{bn} \bar{T}_{abmn} \right\} dx.} \quad (3.5)$$

Thus the relative baseline cross, earlier-root feedback, trace, returned mean, forest/future recoding, and balanced spatial crosses are present once inside one self-adjoint quadratic form. Adding them again as separate functionals would double count the endpoint.

After the Gaussian covariance reduction (3.2), the first derivative contains Q_1 , whose spatial frequencies are $\pm p$ and $\pm 3p$. Stationarity makes $\bar{B}(C_0)$ independent of x , hence Fourier orthogonality gives

$$\mathcal{R}'(0) = 0. \quad (3.6)$$

4. Uniform owner bound

R-130 proves for every frozen real symmetric Q that the physical coefficient energy obeys

$$\left\| \frac{1}{2} D^2(B : Q) \right\|_{\text{op}} \leq H_6 \|Q\|_{\text{op}}, \quad H_6 = \frac{7083}{500P}, \quad P = 4 + 10^{-12}. \quad (4.1)$$

Set

$$a = \|S_1\|_{\text{op}}, \quad b = \|S_2\|_{\text{op}}, \quad d = |p|a, \quad e = 2|p|b. \quad (4.2)$$

For $\|H\|_{\text{HS}} = 1$, (2.8) gives

$$\|C_1\|_{\text{op}} \leq 2ab, \quad \|Q_1\|_{\text{op}} \leq 2de, \quad \|K_1\|_{\text{HS}} := \left(\sum_i \|K_{1,i}\|_{\text{HS}}^2 \right)^{1/2} \leq bd + ae. \quad (4.3)$$

The first term in (3.5) is nonnegative. Since the matrices are six-real, $\|C_1\|_* \leq 6\|C_1\|_{\text{op}}$. Equation (4.1) therefore bounds the mixed covariance term by

$$6H_6(2ab)(2de) = 24H_6abde. \quad (4.4)$$

For the last term take a singular-value decomposition of K_1 , polarize (4.1), and use $\|K_1\|_*^2 \leq 6\|K_1\|_{\text{HS}}^2$. This gives

$$12H_6(bd + ae)^2. \quad (4.5)$$

Substitution of $d = |p|a$, $e = 2|p|b$ yields the exact factor

$$24H_6abde + 12H_6(bd + ae)^2 = 156H_6|p|^2 a^2 b^2. \quad (4.6)$$

If $\lambda_1 = \lambda_{\min} A(p)$ and $\lambda_2 = \lambda_{\min} A(2p)$, the two antipodal signs give

$$a^2 \leq \frac{2}{V\lambda_1}, \quad b^2 \leq \frac{2}{V\lambda_2}. \quad (4.7)$$

After spatial integration, (4.6) becomes

$$\mathcal{R}''(0)[H, H] \geq -\frac{624H_6|p|^2}{V\lambda_1\lambda_2} \|H\|_{\text{HS}}^2. \quad (4.8)$$

5. Exact radial certificate

The exact Sylvester minors of $M - (7/250)I$ are

$$\frac{9}{125}, \quad \frac{1211}{250000}, \quad \frac{89}{31250000}, \quad (5.1)$$

so $M \succ (7/250)I$. Put $x = |p|^2$ and

$$f(x) = x^2 + Zx + r + \frac{7}{250}. \quad (5.2)$$

The exact registered decimals give a negative discriminant, hence $f > 0$. Also $H_6 < 7083/2000$. It remains to prove

$$\frac{624(7083/2000)x}{Vf(x)f(4x)} < \frac{4}{5}. \quad (5.3)$$

For $V = 4096$, clear the positive denominator in (5.3). The resulting quartic is

$$\begin{aligned} P(x) = & \frac{262144}{5}x^4 - \frac{592176264064}{9765625}x^3 \\ & + \frac{74744439754677505969}{1907348632812500}x^2 \\ & - \frac{29970036890042018799}{3051757812500000}x \\ & + \frac{25203778302134079729}{30517578125000000}. \end{aligned} \quad (5.4)$$

Its exact Sturm chain has degrees $(4, 3, 2, 1, 0)$. The signs at zero are $(+, -, -, +, +)$ and the leading signs at positive infinity are $(+, +, -, -, +)$. Both have two variations. Thus P has no positive root; since $P(0) > 0$, (5.3) holds for every $x \geq 0$. Combining (4.8) and (5.3) proves (1.2).

The source cost in this normalized root chart is

$$\frac{9}{20}\mathbb{E}\|tH\xi_1\|^2 = \frac{9}{20}t^2\|H\|_{\text{HS}}^2, \quad (5.5)$$

and therefore has Hessian $9\|H\|_{\text{HS}}^2/10$. Finally, for $R(t) = |w + tz|^2$,

$$\frac{d^2}{dt^2}R(t)^3 = 24R(t)\{(w + tz) \cdot z\}^2 + 6R(t)^2|z|^2 \geq 0. \quad (5.6)$$

Equations (1.2), (5.5), and (5.6) prove (1.3).

6. Owner-coordinate consequence and failed route

For a fixed terminal and filtration, the Doob low and root increments are unique. The R-063 forest and the near/balanced/far Schur blocks are not additional martingale increments. They are alternate formulas or estimates for the same square-minus-trace scalar. The historical R-079 complete-low object is not identified merely by setting a cylinder's primitive low covariance to zero: a returned conditional mean can remain.

Therefore the route that independently fabricates forest, balanced, and low blocks and then adds them to (3.1) is retired. A separate forest/future calculation is legal only as an audit of the already registered identity

$$2P_{\text{comp}} = F_{063}^{\text{ad}} - V_{\text{future}}, \quad (6.1)$$

not as a second payment. This no-go removes an ambiguity in the old production-matrix plan without proving the multi-root estimate.

7. Devil's-advocate audit

1. **Objection: the theorem is only a scalar or same-sign slice. DISMISSED.** H is an arbitrary real 12×12 matrix and mixes both antipodal signs and all six real internal coordinates.
2. **Objection: the primitive trace or future owner was omitted. DISMISSED in the declared chart.** Equation (3.1) starts from the complete covariance-normal endpoint. Gaussian identity (3.2) differentiates that scalar, so trace and all feedback correlations occur exactly once.
3. **Objection: R-130 cannot be used when the frozen matrix Q depends on the state. DISMISSED here.** The matrices C_1, Q_1, K_1 are frozen covariance derivatives at $t = 0$; no state derivative of Q is discarded.
4. **Objection: a numerical 144-dimensional eigenvalue is being reported as proof. DISMISSED.** The sign theorem uses only the exact norm factor (4.6) and rational Sturm certificate (5.4). Quadrature eigenvalues are not part of the theorem statement.
5. **Objection: local positivity closes T-050. UPHELD against that inference.** Nonzero strict-past means, nonlinear h_2 , cross-pair interactions, returned historical low, and the full directed source union are outside this theorem.
6. **Objection: the result selects BCC, a uniform phase, or the current PDE. UPHELD against that inference.** No phase ansatz, vacuum comparison, or candidate-functional discriminator appears here.

External review is invited for the A6/A7 realification factor, simultaneous antipodal normalization, the two Gaussian-IBP factors in (3.2), the half-Hessian convention in (4.1), the nuclear-norm factor six, the SVD polarization factor two, spatial volume, Sturm endpoint signs, and the local-versus-global scope firewall.

8. Result footer and reproduction

Dependencies: A1, A6, A7, R-063, R-079, R-104, R-125, R-128, R-130, R-141, R-149, R-150.

Reproduction command:

```
& E:\Dev\TECT.venv\Scripts\python.exe '
codes/foundations/a13_classii_two_root_endpoint_hessian_uniform_local_gap_boundary_verify.py
```

Result ID: A13-CLASSII-TWO-ROOT-ENDPOINT-HESSIAN-UNIFORM-LOCAL-
GAP-BOUNDARY; Ledger ID: R-151.

Precise statement: at every retained admissible dual-lattice p:2p pair and
 $H \neq 0$, endpoint loss coefficient $< 4/5$, source Hessian $= 9/10$, and local
gap $> 1/10$; the radial certificate holds for every $x \geq 0$.

Scope: zero-past linear 12x12 chart; stationary common-even law; endpoint only.

Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.

Expected output: primary and non-importing independent PASS; integrated PASS.

Falsification gate: any factor, floor, Sturm, scope, hash, PDF, or release check fails.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: no non-lattice/unretained chart, nonzero-past or
nonlinear theorem, multi-root bound, T-050, phase/PDE, or Sector-A closure.

Proof complete: false; T-050 closed: false; Sector A closed: false.

Next required action: lift to nonzero past/nonlinear feedback, then test
owner-complete multi-root aggregation without source reuse.